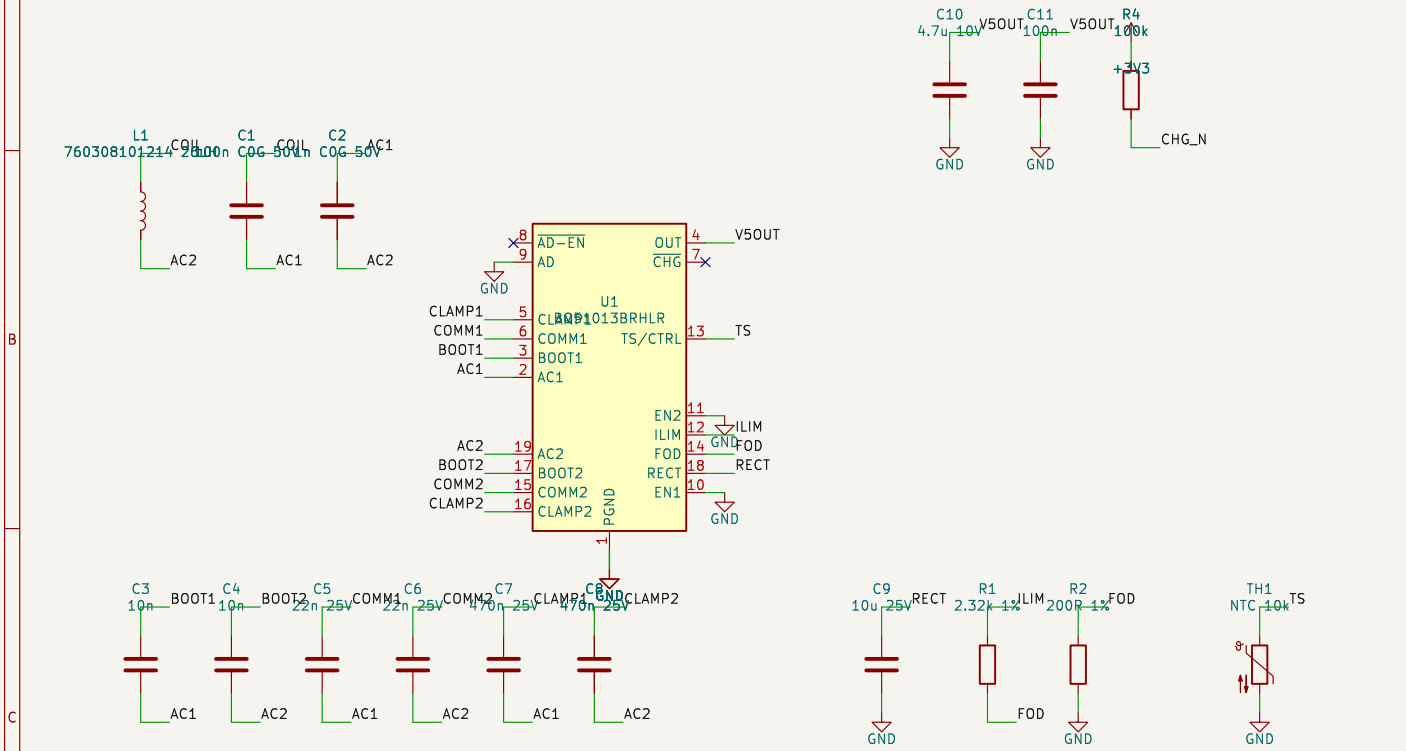
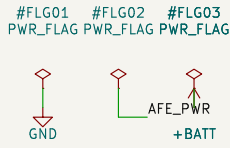
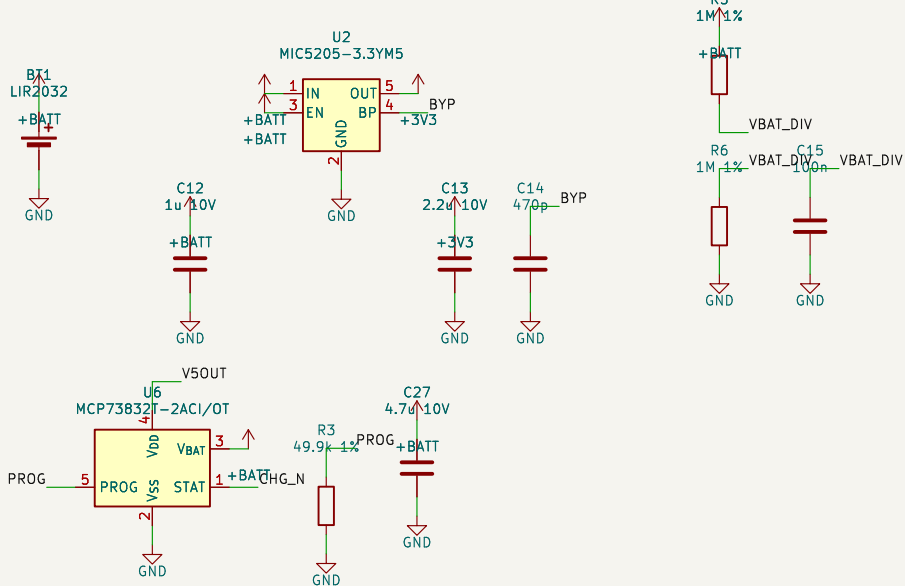


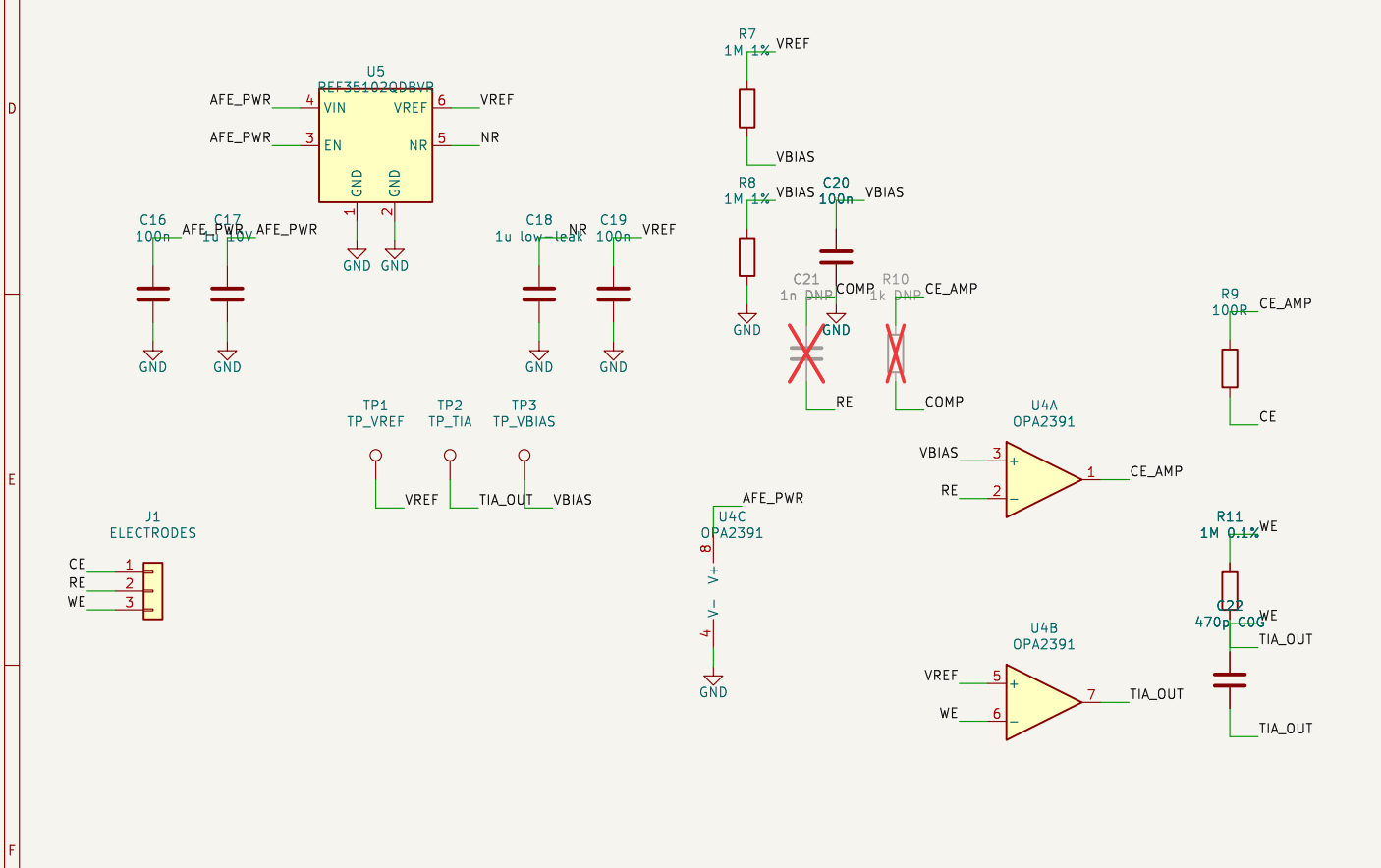
QI RECEIVER 5V (BQ51013B) + 20mA CHARGER (MCP73832)



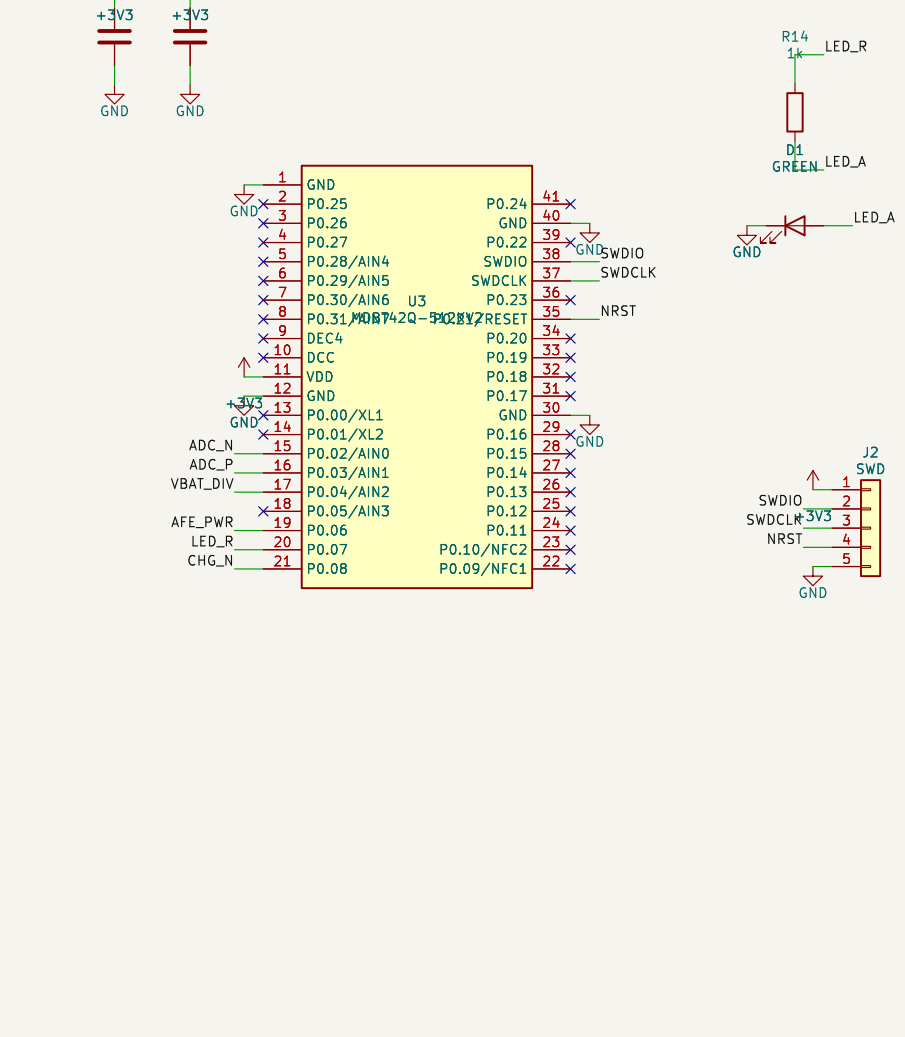
BATTERY + 3.3V LDO + VBAT MONITOR



ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)



nRF52832 MODULE (MDBT42Q) + SWD + LED



DISCRETE POTENTIOSTAT – wireless-charged 3-electrode chronoamperometry  
 $E_{cell} = WE - RE = 1.024V - 0.512V = +0.512V$  fixed.  $I_{WE} = (V(ADC\_P) - V(ADC\_N)) / RF$

- DESIGN NOTES / VERIFY BEFORE LAYOUT
- CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (V5OUT). ILIM: RILIM = 250/IMAX = R1+R2 = 2.52k  $\rightarrow$  IMAX 100mA (HW limit 120mA). MCP73832-2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C). termination -7.5% = 1.5mA, open-drain STAT  $\rightarrow$  CHG\_N (100k pullup to 3V3). This replaces the earlier BQ51050B direct-charge (TI: not recommended <200mA).
  - RESONANT TANK: L1 26uH (Würth 760308101214, dia 19mm = battery-bounded device size). C1 =  $1/(L*(2\pi*100kHz)^2) = 97n \rightarrow 100n$  COG. C2  $\rightarrow$  fd = 1MHz  $\rightarrow$  1n COG. Retune both whenever the coil changes. Small coil = tighter TX alignment; fine at our 0.13W.
  - MDBT42Q pads 22–36 are a sequential reconstruction (P0.09/NFC1 ... P0.23); pads 1–21 and 37–41 verified against the datasheet. Every pad actually used in this design is a verified one. Cross-check the full pad table against the Raytac datasheet before PCB layout.
  - TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (plain 10k = sense defeated, still a valid safe strap). EN1=EN2=GND: wireless enabled.
  - AFE\_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 - 60uA. Power the AFE off between runs to avoid electrode polarization.
  - WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
  - RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (4.27/181nA), 10M/47p (217/92nA).
  - R10/C21 control-amp compensation are DNP: fit only if the CE-RE loop oscillates.
  - LIR2032 has no protection circuit: firmware must System=OFF below 3.0V (VBAT\_DIV, AIN2).
  - SAADC: differential P=AIN1(ADC\_P) N=AIN0(ADC\_N), gain 1/2..4, oversample 64x default.